

DAY-3

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Q1. In plagiarism detection, find the length and the actual Longest Common Subsequence of two documents

Code:

```
def lcs(s1, s2):
    m = len(s1)
    n = len(s2)

    dp = [[0] * (n + 1) for _ in range(m + 1)]

    # Build DP table
    for i in range(m):
        for j in range(n):
            if s1[i] == s2[j]:
                dp[i + 1][j + 1] = dp[i][j] + 1
            else:
                dp[i + 1][j + 1] = max(dp[i][j + 1], dp[i + 1][j])

    # Find actual LCS
    i, j = m, n
    lcs_string = ""

    while i > 0 and j > 0:
        if s1[i - 1] == s2[j - 1]:
            lcs_string = s1[i - 1] + lcs_string
            i -= 1
            j -= 1
        elif dp[i - 1][j] > dp[i][j - 1]:
            i -= 1
        else:
            j -= 1
```

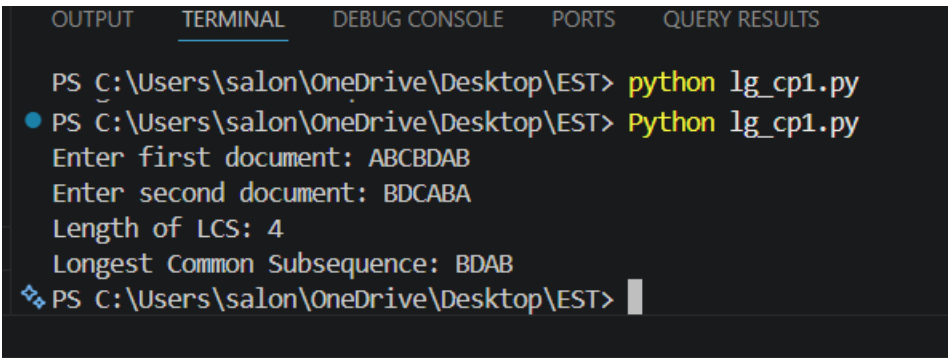
```
print("Length of LCS:", len(lcs_string))  
print("Longest Common Subsequence:", lcs_string)
```

Input

```
doc1 = input("Enter first document: ")  
doc2 = input("Enter second document: ")
```

```
lcs(doc1, doc2)
```

Output:



```
OUTPUT  TERMINAL  DEBUG CONSOLE  PORTS  QUERY RESULTS  
  
PS C:\Users\salon\OneDrive\Desktop\EST> python lg_cp1.py  
● PS C:\Users\salon\OneDrive\Desktop\EST> Python lg_cp1.py  
Enter first document: ABCBDAB  
Enter second document: BDCABA  
Length of LCS: 4  
Longest Common Subsequence: BDAB  
❖ PS C:\Users\salon\OneDrive\Desktop\EST>
```

Q2. Given a city network, compute the shortest distances between all pairs of cities.

Code:

```
INF = 99999
```

```
graph = [  
    [0, 5, INF, 10],  
    [INF, 0, 3, INF],  
    [INF, INF, 0, 1],  
    [INF, INF, INF, 0]  
]
```

```
n = len(graph)
```

```
for k in range(n):  
    for i in range(n):  
        for j in range(n):  
            graph[i][j] = min(graph[i][j], graph[i][k] + graph[k][j])
```

```
print("Shortest distances between all pairs of cities:")
```

```
for i in range(n):  
    for j in range(n):  
        if graph[i][j] == INF:  
            print("INF", end="\t")  
        else:  
            print(graph[i][j], end="\t")  
    print()
```

Output:

```
PS C:\Users\salon\OneDrive\Desktop\EST> python lg_cp1.py  
Shortest distances between all pairs of cities:  
0      5      8      9  
INF    0      3      4  
INF    INF    0      1  
INF    INF    INF    0  
PS C:\Users\salon\OneDrive\Desktop\EST>
```